U24AI049

Assignment 8

Use the following graph for the Traveling Salesperson problem. Find the least cost path in the graph for the TSP problem.

	A	B	C	D	E	F	G	H	
A	0,	10,	15,	20,	25,	30,	35,	40],	# City 0
B	12,	0,	35,	15,	20,	25,	30,	45],	# City 1
C	25,	30,	0,	10,	40,	20,	15,	35],	# City 2
D	18,	25,	12,	0,	15,	30,	20,	10],	# City 3
E	22,	18,	28,	20,	0,	15,	25,	30],	# City 4
F	35,	22,	18,	28,	12,	0,	40,	20],	# City 5
G	30,	35,	22,	18,	28,	32,	0,	15],	# City 6
H	40,	28,	35,	22,	18,	25,	12,	0]	# City 7

1. Use the local beam search to solve the problem. Give a comparative analysis of the algorithm when the beam width k=3,5,10. Does the convergence depend on the value of k?

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  COMPARATIVE ANALYSIS
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eration, so the
    search escapes local optima more effectively.
  - k=3 is fastest per run but risks getting stuck early.
  - k=10 explores more broadly and finds lower-cost tours.
  - Yes, convergence quality DOES depend on k.


Shedha gami@LAPTOP-8UU5LAV5 MINGW64 ~/AI202 lab/Assignment 8
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2. Use the genetic algorithm to solve the problem. Use one crossover point and two crossover points to generate the offspring. Does the number of crossover points impact the convergence rate?

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  - Yes, convergence quality DOES depend on k.
eration, so the
    search escapes local optima more effectively. …

  >>> Best Path : B -> A -> C -> D -> H -> G -> F -> E -> B
  >>> Total Cost: 121

  --- Comparative Analysis ---
  1-point crossover → Cost: 125
  2-point crossover → Cost: 121

  Conclusion:
  2-point crossover preserves more contiguous subtours from both
  parents, producing more diverse offspring. 1-point crossover
  keeps a larger intact prefix from one parent, which can help or
  hurt depending on population structure. In practice, 2-point OX
  often converges to better solutions for TSP due to higher diversity.
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Codes Github and local computer me hai as they are too long